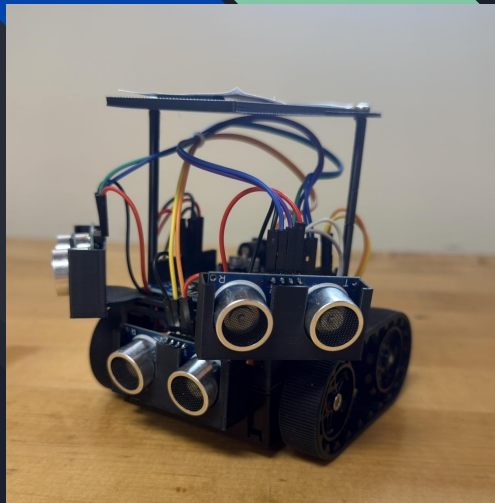


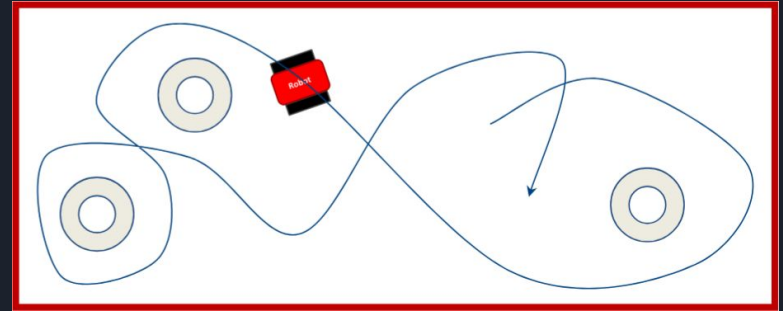


Autonomous Robot Project

By: Abilash Ramasamy, Aryan Bhatnagar, and Jaiden Garcia

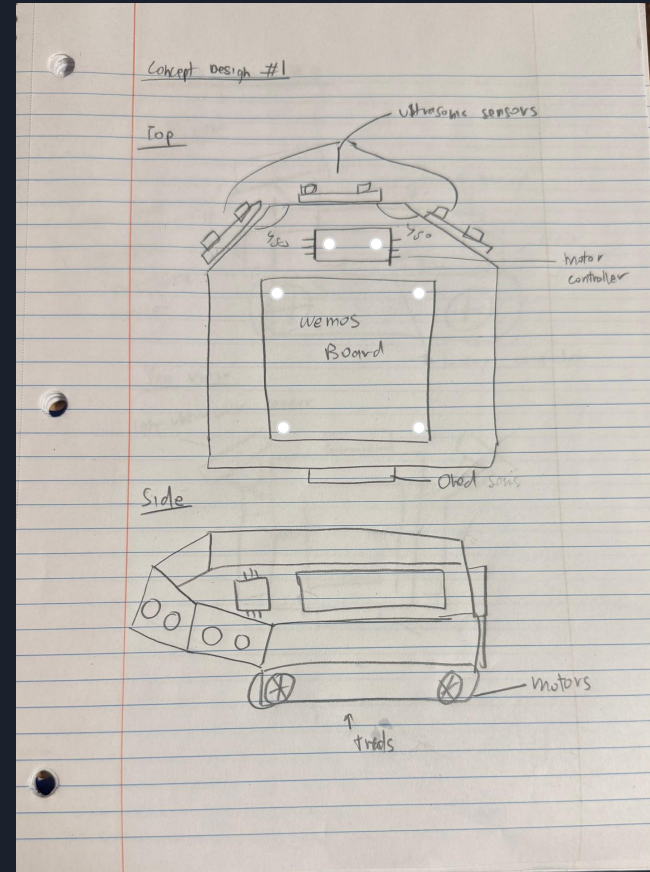
Problem Definition

- Real-time obstacle avoidance to reach targets in time
- Constraints
 - 5 Hour print time
 - Fixed Components
 - No breadboard + Organized Power Bus



Concept Development - #1

- Placed all three Ultrasonic sensors in the front
- Secure the motor controller placed horizontally across the robot
- Secure the WEMOS board running vertically down the robot

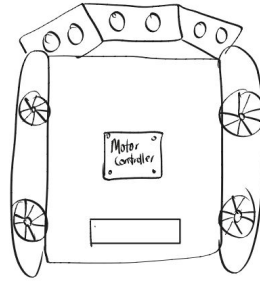


Concept Development - #2

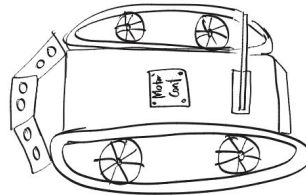
- Kept the original positioning of the ultrasonic sensors
- Integrated a vertical mount for the WEMOS board
- Holes present on the mount to screw in the board

Concept Design #2-

Top View



Side View

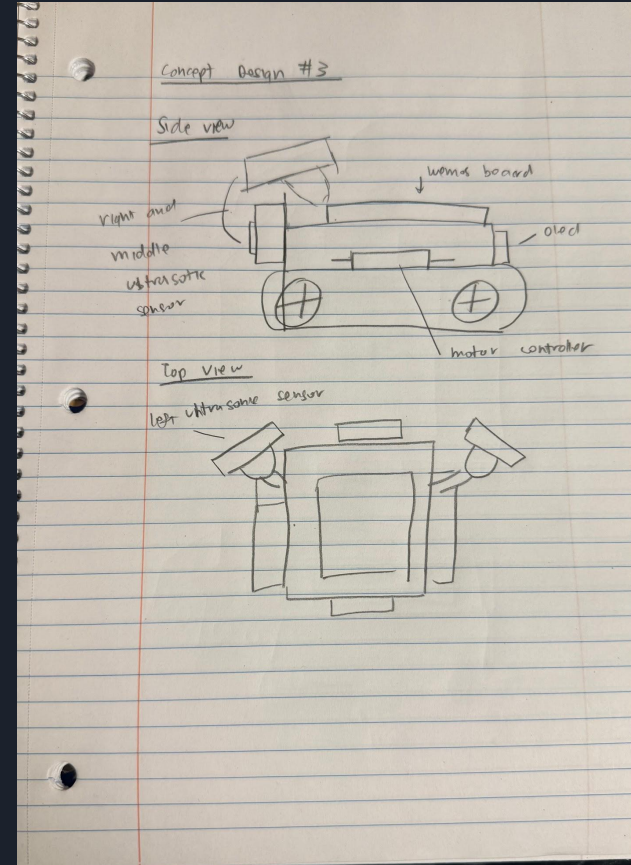


Back View



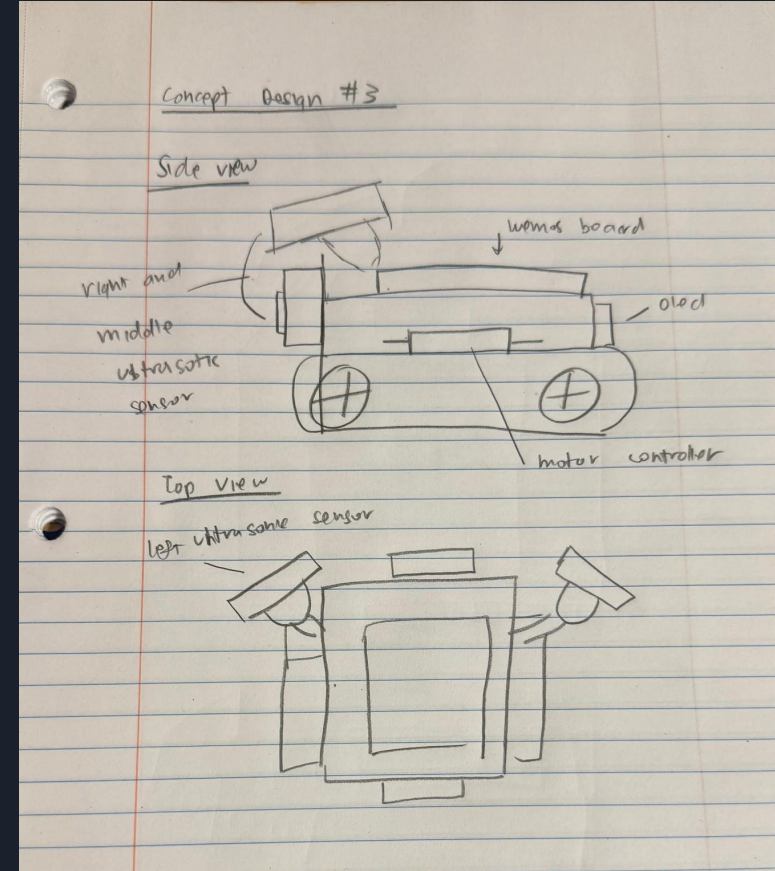
Concept Development - #3

- Base screwed into robot —> holds motor controller
- Separate part for the L & R Ultrasonic sensors —> holds WEMOS board
- Elevation of left and right Ultrasonic sensors



Concept Chosen - Concept #3

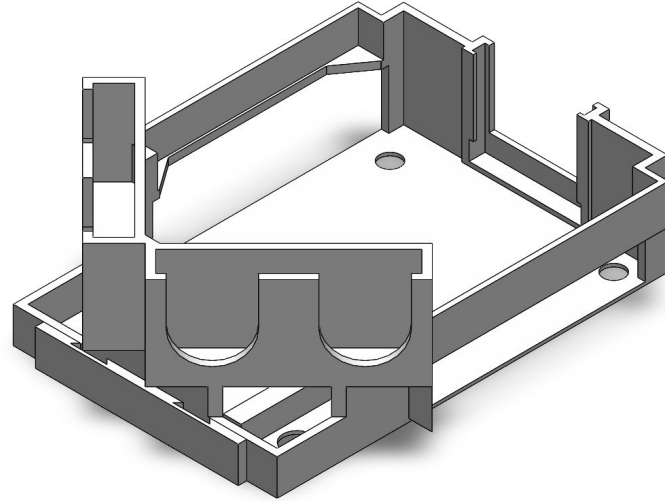
Criteria	Concept #1	Concept #2	Concept #3
Size	3	1	2
Durability	3	2	1
Design	3	2	1
Easy to make	1	2	3
Safety	3	2	1
Total Score (Lowest Wins)	13	9	8



Final Design

Design #1

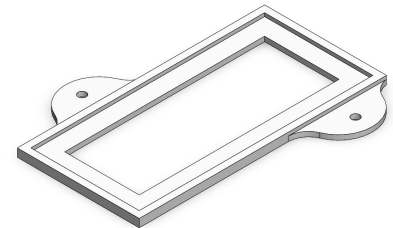
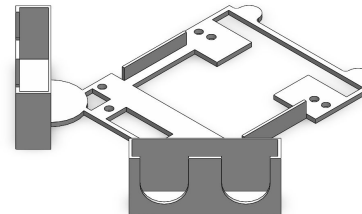
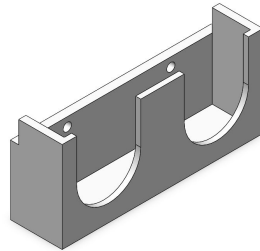
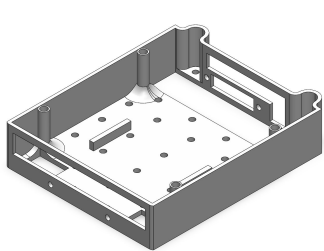
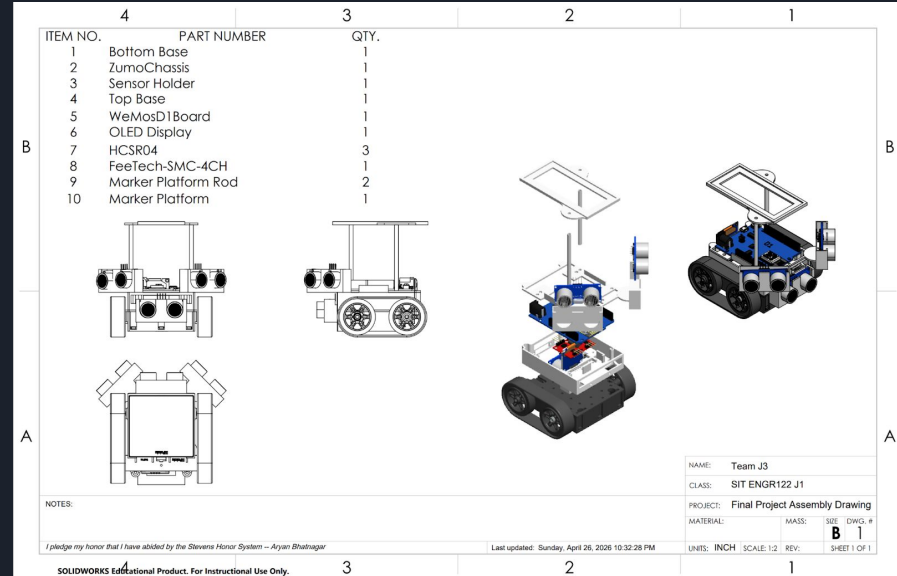
- Original design allowed for two Ultrasonic sensors and OLED display
- Enough to accomplish Back Forth Road Test and Left Hand Loop
- Extravagant printing time + inefficient

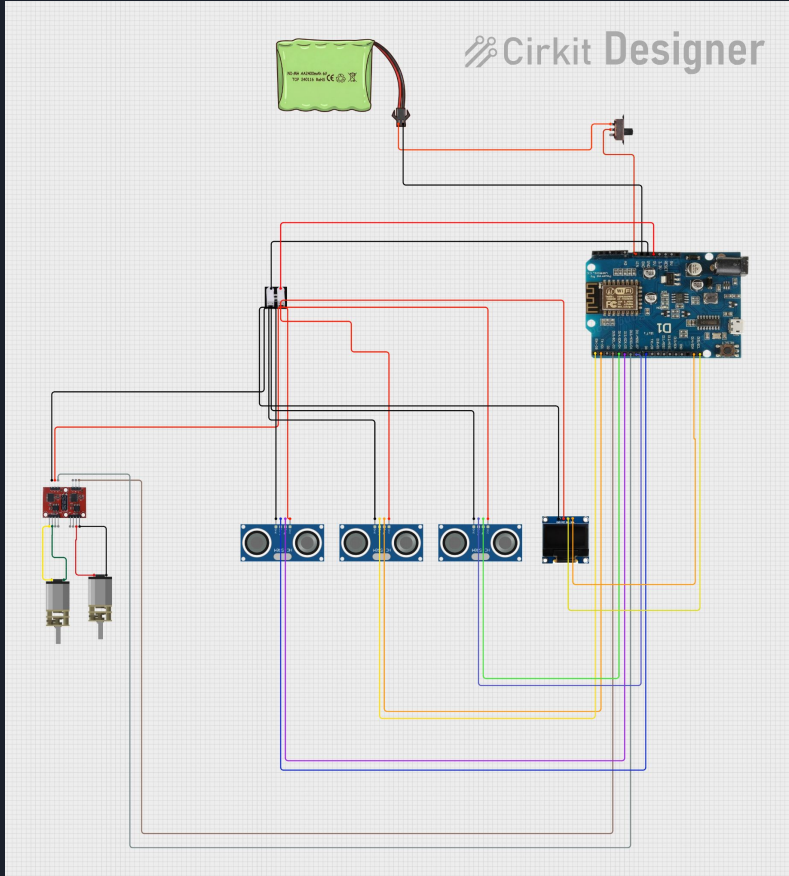


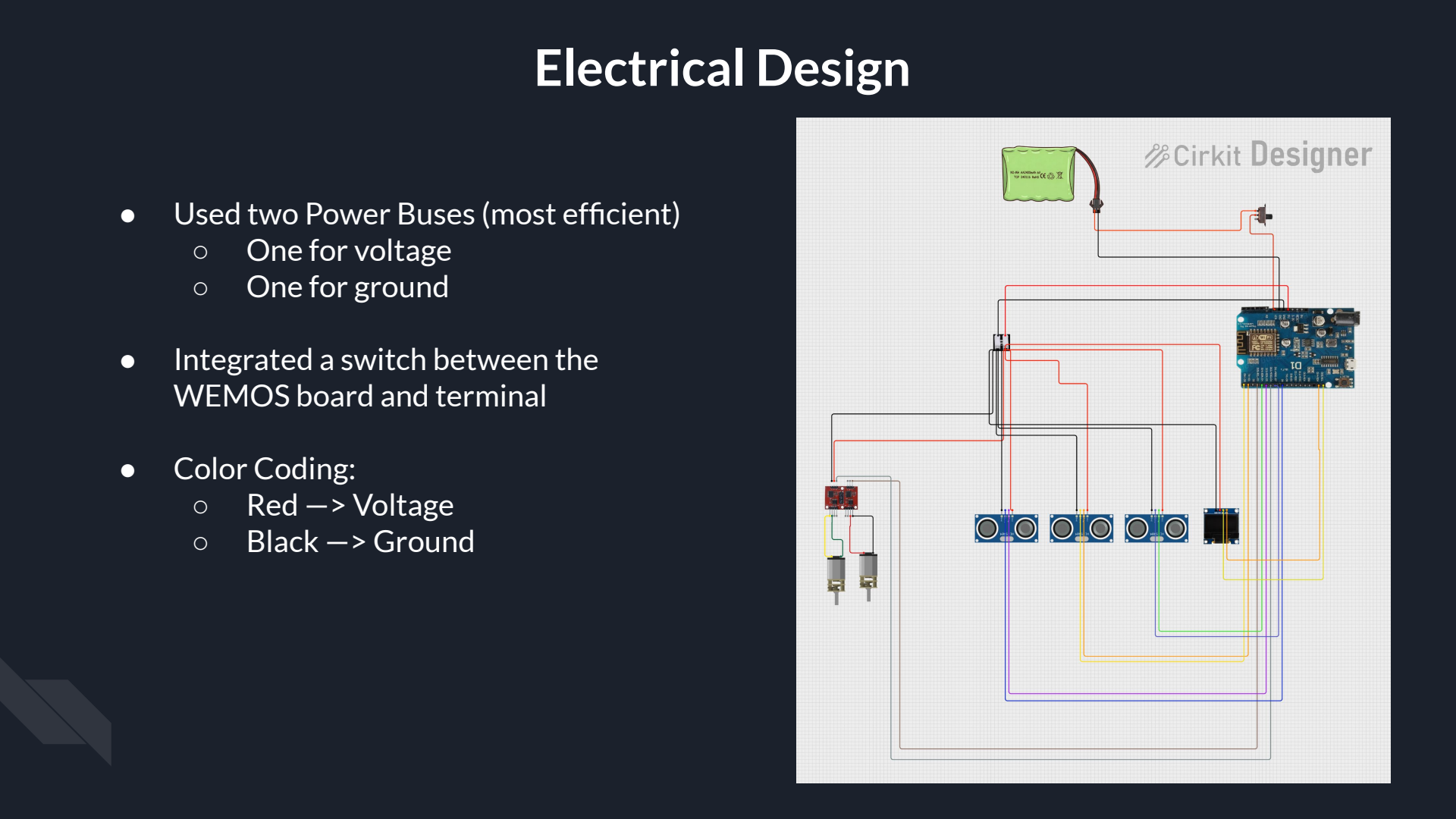
Final Design

Design #2

- Three different parts:
- Much more efficient design —> utilized 3 Ultrasonic sensors
- Was able to stay well under printing time limit

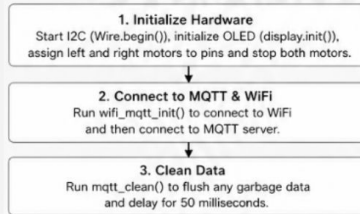


- # Electrical Design
- Used two Power Buses (most efficient)
 - One for voltage
 - One for ground
 - Integrated a switch between the WEMOS board and terminal
 - Color Coding:
 - Red —> Voltage
 - Black —> Ground
- 

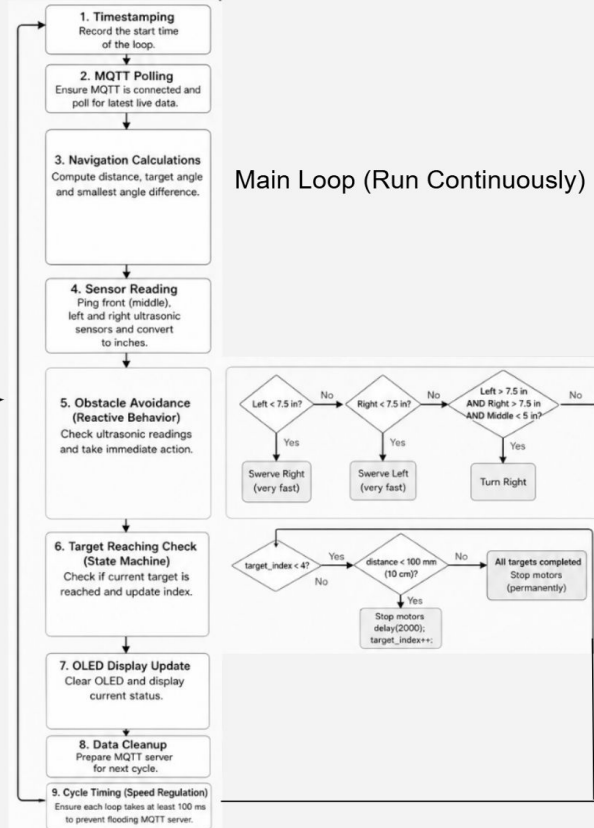


Software Algorithms

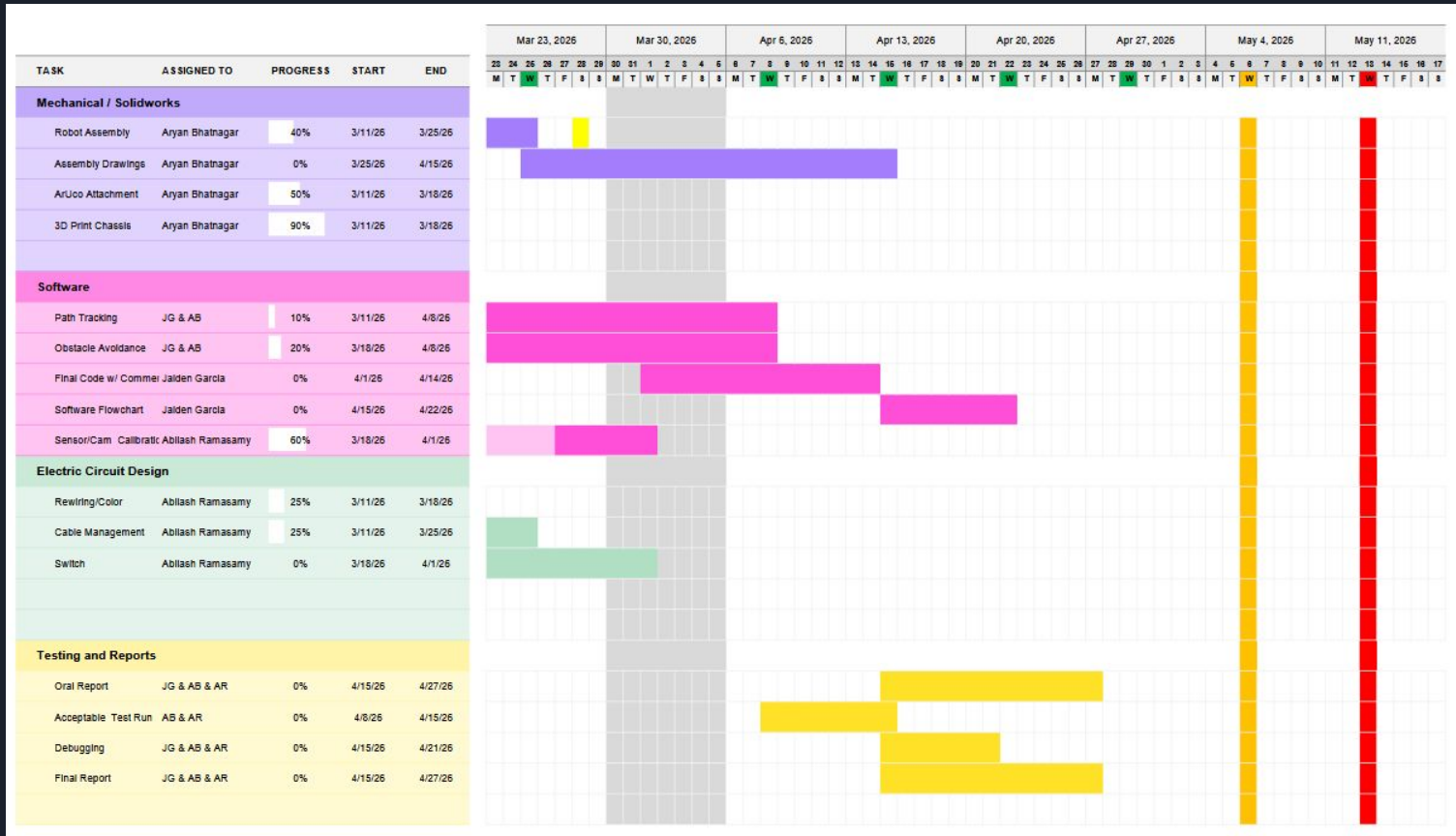
Setup and Initialize (run once)



Main Loop (Run Continuously)

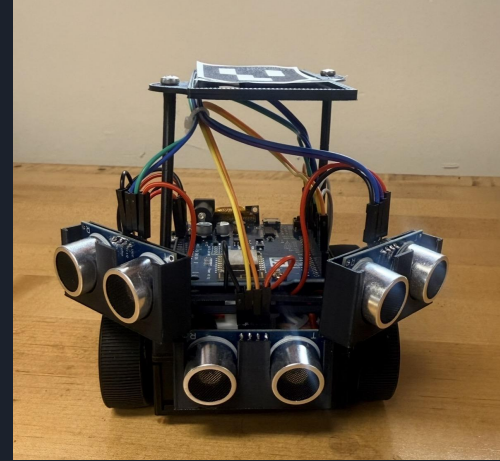


Project Management



Results of Week 13 testing

- Initial run time of 1 minute, qualifying for pre-screening with 207.5 points
- Reduced the execution time to 51 seconds (214.5 points), flawless execution



The following section *must* be completed by your instructor and pre-screening test supervisor.

Professor: Shigano Date: 4/22/2026

Test Supervisor (TA): Alexander Stangor [Signature] Date: 4/22/2026

Test Scores :

- 1st Trial Score: <u>210</u>	Date&Time: <u>4/24/26 @ 3:00</u>	Signature: <u>[Signature]</u>
- 2nd Trial Score: <u>214.5</u>	Date&Time: <u>4/24/26 @ 4:33</u>	Signature: <u>[Signature]</u>
- 3rd Trial Score: <u>213.1</u>	Date&Time: <u>4/24/26 @ 5:05</u>	Signature: <u>[Signature]</u>

Conclusion & Recommendations

- Robot successfully navigated to all targets in 51 seconds without collisions.
- Performance improved with constant iteration
- Possible recommendations :
 - ArUco Stand
 - Tread Alignment
 - Fully Enclosed

